

STAT 314500 - Applied Partial Differential Equations
Assignment 2: Separation of variables and Fourier Series

This is an individual assignment. You may collaborate with others but are expected to submit your own work. Show all relevant work. Incomplete work will be docked points.

Note: if a question is marked with an asterisk, *, and appears vague - that's on purpose. Questions marked * are conceptual short-answer questions. Do your best to reason these through, and describe that reasoning concisely. We are looking more for reasoning than precision.

Reading: Chapters 2 and 3 in Applied Partial Differential Equations with Fourier Series and Boundary Value Problems (Richard Haberman).

Question 1

1. What ODEs do one obtain when solving the following PDE using the method of separation of variables? (5 points)

$$\frac{\partial u}{\partial t} = \frac{k}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right)$$

2. For the differential equation (5 points)

$$\frac{d^2 \phi}{dx^2} + \lambda \phi = 0, \quad x \in (0, L),$$

determine the eigenvalues λ and the corresponding eigenfunctions if ϕ satisfies the boundary condition $\frac{d\phi}{dx}(0) = 0, \phi(L) = 0$. In particular assuming λ is real valued, analyze the three cases $\lambda = 0, \lambda > 0$ and $\lambda < 0$.

3. Solve the heat equation (5 points)

$$\begin{cases} \frac{\partial u}{\partial t} - k \frac{\partial^2 u}{\partial x^2} = 0 \\ u(x, 0) = 3 \sin \frac{\pi x}{L} - \sin \frac{3\pi x}{L} \\ u(0, t) = 0 \\ u(L, t) = 0. \end{cases}$$

Question 2

Let $x \in [0, L]$. Consider the boundary value problem

$$\begin{cases} \frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} \\ \frac{\partial u}{\partial x}(0, t) = 0 \\ \frac{\partial u}{\partial x}(L, t) = 0 \\ u(x, 0) = f(x) \end{cases}$$

1. Solve the PDE above using separation of variables. [Hint: The answer is of the form $u(x, t) = A_0 + \sum_{n=1}^{\infty} A_n e^{-\lambda_n k t} \cos \frac{n\pi x}{L}$] (5 points)
2. Show that the initial condition is satisfied if $f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi x}{L}$. Solve for A_0 and A_n for $n \geq 1$. (5 points)

3. What happens to the temperature distribution as $t \rightarrow \infty$? How does this compare with your solution from the previous homework? (5 points)

Question 3 Find and sketch the Fourier series of the following function in the interval $x \in (-L, L]$: (10 points)

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Question 4 Find and sketch the Fourier cosine series of $f(x) = \sin \frac{\pi x}{L}$ in $x \in [0, L]$. Briefly discuss. [Hint: What is the even extension of f ?] (10 points)